

1 Introduction

In my previous paper, I built a simple Bayesian game with a single consumer. This paper extends that setting to a small niche market with heterogeneous consumers each with a different willingness to pay. The producer does not make any parametric assumption about the distribution of consumers' willingness to pay (except regularity conditions, e.g. an atomless F with a density), but only seeks the optimal quantile that maximise the profits. This yields a general Bayesian game well-suited to a DQN. While a fully non-parametric approach can be intractable when maintaining the entire posterior over distribution, I argue that two statistics - the optimal price and the local hazard rate - characterise the producer's learning, making the model tractable.

2 Theoretical Base

In this game, we have N heterogeneous consumers, each with private willingness to pay W_i . Each consumer's policy is mixed :

$$\epsilon_i \mathbb{1}(W_i \geq x_t) + \frac{(1 - \epsilon_i)}{2}, \quad \epsilon_i \in [0, 1]$$

where ϵ_i controls the level of obfuscation. Consumers try to learn optimal ϵ_i^* via DQN.

The producer is single and updates a Bayesian belief about consumer willingness to pay. Rather than tracking the entire posterior over distributions, the producer maintains only two statistics: optimal price x^* and local hazard rate h^* at x^* . The motivation for this dimension reduction is statistical : with N consumers the producer observes an empirical CDF $\hat{F}_N$ satisfying $\sup_x |\hat{F}_N(x) - F(x)| = O_p(N^{-1/2})$ by the Dvoretzky-Kiefer-Wolfowitz inequality. Under the atomless assumption, the plug-in estimates $(\hat{x}^*, \hat{h})$ are consistent, so a market with sufficiently large N lets the producer recover the two-dimensional statistic with high probability. The producer's expected profit at price x , given aggregate obfuscation $\bar{\epsilon}$, is

$$\Pi(x; \bar{\epsilon}) = (x - c)\tilde{S}(x; \bar{\epsilon}), \quad \tilde{S}(x; \bar{\epsilon}) := \bar{\epsilon}(1 - F(x)) + \frac{1 - \bar{\epsilon}}{2}$$

where c is the unit production cost, $\tilde{S}$ is the effective demand. The constant term $\frac{1 - \bar{\epsilon}}{2}$ is a price insensitive noise floor of acceptances produced by randomisation.

2.1 Reduction of Dimensionality

Lemma 2.1. *Assume F is differentiable with density f and $\tilde{f}(x) = \bar{\epsilon}f(x) > 0$ at any interior maximiser. Then every interior optimal price x^* satisfies*

$$x^* = c + \frac{\tilde{S}(x^*; \bar{\epsilon})}{\tilde{f}(x^*)} = c + \frac{1}{\tilde{h}(x^*; \bar{\epsilon})}$$

pf. $\Pi(x; \bar{\epsilon}) = (x - c)\tilde{S}(x; \bar{\epsilon})$ is differentiable with $\Pi'(x) = \tilde{S}(x) - (x - c)\tilde{f}(x)$. At an interior maximiser $\Pi'(x^*) = 0$; dividing by $\tilde{f}(x^*) > 0$ gives $x^* - c = \tilde{S}(x^*)/\tilde{f}(x^*) = 1/\tilde{h}(x^*)$. Since $\tilde{S}(x; \bar{\epsilon}) \rightarrow \frac{1-\bar{\epsilon}}{2} > 0$ as x grows, the noise floor can push the maximiser to the upper boundary of the price domain; an interior optimum need not exist for $\bar{\epsilon} < 1$. This distortion relative to the truthful benchmark ($\bar{\epsilon} = 1$) is the mechanism by which obfuscation operates. We therefore work on a bounded price domain $x \in [0, \bar{x}]$ on which the lemma characterises any interior optimum and Kakutani gives existence.

Theorem 2.1 (Sufficient Statistic). *The producer's local optimal pricing depends on F only through the local hazard function near x^* .*

pf. By the previous lemma, any interior optimum solves $x = c + 1/\tilde{h}(x; \bar{\epsilon})$, which involves F only through $\tilde{h}(x; \bar{\epsilon}) = \bar{\epsilon}f(x)/\tilde{S}(x; \bar{\epsilon})$ evaluated near x . Hence two type distributions inducing the same effective hazard on a neighbourhood of their optimum share the same interior optimal price; the optimality check depends on F only through the local pair $(x, \tilde{h}(x; \bar{\epsilon}))$.

→the producer's learning state can be taken to be the two-dimensional $(x_t, \tilde{h}_t)$ which we use as the input to the DQN, paralleling the (μ, σ) state of the single-type model.

2.2 Existence of Equilibrium in the Game

Assume F , the CDF of consumer willingness-to-pay, is atomless. Then the expected demand $\tilde{S}$ is a continuous function of $(x, \bar{\epsilon})$ even though individual accept indicators jump - the discontinuity gets integrated out over the type distribution F . By Berge's maximum theorem, the producer's best-response correspondence $X^*(\bar{\epsilon}) = \arg\max_x \Pi(x; \bar{\epsilon})$ is nonempty-valued and u.h.c.; the induced aggregate best response map on $\bar{\epsilon} \in [0, 1]$ is u.h.c. with nonempty, convex (from mixed obfuscation strategies), compact values. Therefore by the Kakutani's fixed point theorem, the fixed point (Bayes-Nash equilibrium) exists.